Cloud Cost Optimization Plan

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Abstract

This Cloud Cost Optimization Plan addresses a mid-sized e-commerce company's 150% increase in cloud spending compared to a 75% rise in revenue. The objective is to reduce costs while maintaining performance during peak shopping periods. The plan includes analyzing current resource utilization, implementing auto-scaling, rightsizing underused instances, optimizing storage tiers and databases, and improving CDN efficiency. Continuous monitoring through tools like AWS Cost Explorer and CloudWatch will ensure sustained cost efficiency without impacting availability or customer experience.

1) Analysis of resource utilization and cost optimization:

Although the e-commerce business is growing rapidly,
revenue is not increasing proportionally due to rapidly rising cloud costs. The table below presents an analysis of resource utilization along with potential cost optimization opportunities.

Resource Type	Utilization	Analysis	Optimization Feedback
Web Servers (50 total)	50 running continuously	Headroom isn't being used; cloud resources should adjust based on real-time demand (auto-scaling)	Servers are statically allocated based on actual demand instead of scaling dynamically.
Database Servers	40% average utilization	Low utilization indicates over-provisioning	Over-provisioned; storage allocated is larger and more expensive than required.
Storage (500 TB total)	500 TB mixed access	Paying unnecessarily high rates for storing rarely used data	move infrequently accessed data to cheaper storage classes
CDN	Global	Usage likely spikes during promotions	Optimize caching policies to reduce origin fetches.
Backup & DR Systems	Active	Match data storage needs with cost-effective solutions	Backup retention and storage tiering are misaligned with business needs, causing extra costs.

2) Cloud cost optimization tools :

If an e-commerce company adopts AWS, analyzing spending patterns and usage trends should begin with AWS Cost Explorer. This service helps track and visualize cost data over time, identify which resources drive the most expenses, and uncover usage patterns or anomalies.

For optimizing compute resources, AWS Compute Optimizer should be used to receive rightsizing recommendations. It evaluates the performance metrics of EC2 instances, RDS databases, and Lambda functions, then suggests optimal instance types that balance performance with cost efficiency.

To detect inefficiencies in the environment, such as idle resources, over-provisioned instances, and storage optimization opportunities, the company should rely on AWS Trusted Advisor. It provides actionable insights to reduce waste and improve cost-effectiveness across multiple AWS services.

Finally, to predict future workloads and reduce costs for steady usage patterns, the company should consider AWS Savings Plans and Reserved Instances. These options allow for committing to a consistent level of usage in exchange for significant discounts compared to on-demand pricing.

Together, these AWS tools provide a complete framework for continuous cost monitoring, optimization, and forecasting, ensuring the e-commerce platform maintains performance while controlling cloud expenditure.

3) Recommendations for right-sizing, reserved instances, auto-scaling, and storage optimization

Web Servers

- Implement auto-scaling to handle real-time transaction based on CPU size, memory, and request metrics.
- During off-peak time, moving normal workloads to spot instances .
- For cost reduction evaluate reserved instances for primary loads.

Databases

- Analyze utilization patterns: move ideal instances to smaller sizes.
- Consider server-less or auto-scaling database options. And use reserved instances for predictable workloads.

Storage

- Implement lifecycle policies to move older or less frequently accessed data to archival storage such as S3 glacier. Enable data de-duplication and compression to optimize storage.

CDN & Backup

- Optimize caching policies to reduce origin server calls.
- Decrease inter-region data transfer costs. Review backup schedules and retention; consider incremental backups over full daily backups.

4) Timeline with risk assessment

Phase 1: Assessment & Tool Setup (Week 1-2)

- Enable Cost Explorer/Advisor, analyze usage
- Minimal risk; only monitoring setup

Phase 2: Web & DB Optimization (Week 3-5)

- Implement auto-scaling, right-size Data Base, reserved instances.
- Risk of performance impact; mitigate with staged roll-out & testing.

Phase 3: Storage Optimization(Week 6-7)

- Tiering and lifecycle policies
- Data retrieval delays; mitigate with access pattern review.

Phase 4: Backup & DR Optimization(Week 8)

- Optimize backup frequency & retention
- Slightly longer recovery time; ensure SLAs are met

Phase 5: Monitoring & Continuous Improvement(Week 9 onward)

- Setup dashboards and alerts when its heating spike.
- Low risk; ongoing monitoring

5) Monitoring framework to track optimization results:

Proper monitoring framework is also necessary to ensure visibility into cloud resource usage, performance, and costs. It helps identify inefficiencies early, maintain optimal system performance, and prevent unexpected cost spikes. Below are some of key factors need to be track and analyze properly.

- Measure the actual CPU, memory, and I/O utilization of EC2 instances and RDS databases against their allocated capacity. This helps identify rightsizing actions to perform task and reduce unnecessary costs.
- Monitor storage consumption across different tiers such as S3 Standard and Glacier. As per data access frequency and lifecycle patterns confirms that data is stored in the most cost-effective tier without compromising availability or performance.

- Track content delivery metrics such as the number of requests served from the origin versus those served from the cache. A high cache hit ratio indicates efficient content distribution, reducing load on origin servers and lowering data transfer costs.
- Evaluate how much storage is being consumed by backups and replicated data across regions. Regularly review retention policies and replication settings to balance redundancy requirements with cost efficiency.
- Continuously compare actual cloud spending against forecaste budgets and usage projections. This allows early detection of deviations, supports proactive cost control, and ensures alignment with financial planning.

6) Cost savings and ROI:

The e-commerce company's current expenditure on its cloud environment stands at approximately $16,163.95 per month, which translates to an annual total of nearly $193,967.40.

This monthly spend is distributed across key services:

- Web servers: $3,697.22
- Database servers: $1,630.14
- Object storage: $3,822.00
- Content Delivery Network (CDN): $870.00
- Backup and disaster recovery (DR) systems: $6,144.00

Costs are primarily driven by three inefficiencies: continuously running web servers, underutilized database instances, and the use of expensive storage tiers for data that is infrequently accessed. Optimization efforts will strategically target these areas of inefficiency:

1. Web Servers:Servers will be migrated to an auto-scaling model, ensuring capacity dynamically expands during peak demand and contracts during low traffic. Combined with the utilization of Reserved and Spot Instances, this is projected to reduce web server costs by approximately 35%, yielding a monthly saving of $1,294.02.
2. Databases:Databases, which average only 40% utilization, will be right-sized and partially transitioned to serverless architecture or smaller instance types. This change is expected to cut database costs by 30%, saving roughly $489.00 monthly.
3. Storage:Storage expenses are expected to decline by approximately 40% ($1,528.80) by implementing lifecycle management policies. These policies will automatically transfer older or rarely accessed data to Amazon S3 Glacier.
4. CDN:CDN efficiency will be improved by refining caching policies to increase the cache hit ratio, thereby reducing data transfer costs by 20% (i.e., $174.00).

aws

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Estimate url

Estimate summary

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	16,163.95 USD	193,967.40 USD Includes upfront cost

Detailed Estimate

Name	Group	Region	Upfront cost	Monthly cost
Amazon EC2	-	US East (N. Virginia)	0.00 USD	2,591.50 USD

Status -

Description: -

Config summary Tenancy (Shared Instances), Operating system (Linux), Workload (Consistent, Number of instances: 50), Advance EC2 instance (m5.large), Pricing strategy (Compute Savings Plans 1yr No Upfront), Enable monitoring (disabled), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month)

Name	Group	Region	Upfront cost	Monthly cost
Amazon EC2	-	US East (N. Virginia)	0.00 USD	1,105.72 USD

Status -

Description: -

Config summary Tenancy (Shared Instances), Operating system (Linux), Workload (Monthly, Baseline: 20, Peak: 30, Duration of peak: 3 Day 0 Hr 0 Min), Advance EC2 instance (m5.large), Pricing strategy (Compute Savings Plans 1yr No Upfront), Enable monitoring (disabled), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month)

Name	Group	Region	Upfront cost	Monthly cost
Amazon RDS for MySQL	-	US East (N. Virginia)	0.00 USD	1,630.14 USD

Status -

Description: -

Config summary Storage amount (100 GB), Storage for each RDS instance (General Purpose SSD (gp2)), Nodes (10), Instance type (db.m5.large), Utilization (On-Demand only) (40 %Utilized/Month), Deployment option (Multi-AZ), Pricing strategy (OnDemand)

Name	Group	Region	Upfront cost	Monthly cost
Amazon Simple Storage Service (S3)	-	US East (N. Virginia)	0.00 USD	3,822.19 USD

Status -

Description: -

Config summary S3 Standard storage (150 TB per month), Data returned by S3 Select (45 TB per month) S3 Glacier Flexible Retrieval Average Object Size (16 MB), Provisioned Capacity Units (0), null (0) S3 Glacier Deep Archive Average Object Size (16 MB), S3 Glacier Deep Archive storage (350 TB per month)

Name	Group	Region	Upfront cost	Monthly cost
Amazon CloudFront	-	US East (N. Virginia)	0.00 USD	870.40 USD

Status -

Description: -

Config summary Data transfer out to internet (10 TB per month)

Name	Group	Region	Upfront cost	Monthly cost
AWS Backup	-	US East (N. Virginia)	0.00 USD	6,144.00 USD

Status -

Description: -

Config summary Estimated daily change of primary data (%) (0.0056), Estimated annual increase in primary data (%) (0.4), Amount of primary data to be backed up (100 TB)

Name	Group	Region	Upfront cost	Monthly cost
AWS Data Transfer	-	US East (N. Virginia)	0.00 USD	0.00 USD

Status -

Description: -

5. Backup/DR:Backup and disaster recovery systems will adopt incremental backups and align retention periods strictly with business requirements, lowering that specific expense by 25% ($1,536.00).

Following full implementation, monthly cloud costs are projected to drop from $16,163.95 to around $11,142.13, generating substantial annual savings of $133,705.00.

Implementation, which includes configuration, monitoring setup, and testing, requires a one-time investment of approximately $18,000.00.

The resulting Return on Investment (ROI) is calculated as:
ROI= (Total Annual Savings -Investment)/Investment *100= (193,967.00 -15,000) /15,000*100=977.59%

This exceptional ROI indicates that every dollar invested in optimization yields $9.77 in yearly savings, resulting in a payback period of just over one month.

Beyond direct cost reduction, this initiative enhances scalability, operational efficiency, and system visibility. Continuous monitoring via AWS Cost Explorer, Compute Optimizer, and Trusted Advisor will ensure sustained optimization, allowing the company to maintain high performance while effectively controlling cloud expenditure.

Reference

a) https://docs.aws.amazon.com/cost-management/latest/userguide/ce-rightsizing.html

b) https://docs.aws.amazon.com/compute-optimizer/latest/ug/savings-estimation-mode.html

c) https://aws.amazon.com/blogs/aws-cloud-financial-management/launch-aws-cost-explorer-rightsizing-recommendations-integrates-with-aws-compute-optimizer/

d) https://docs.aws.amazon.com/AmazonS3/latest/userguide/cost-optimization.html

e) https://www.lucidity.cloud/blog/cloud-cost-optimization-best-practices

f) https://www.sedai.io/blog/strategies-to-improve-cloud-efficiency-and-optimize-resource-allocation

g) https://learn.microsoft.com/en-us/azure/well-architected/cost-optimization/optimize-scaling-costs